# Attention Is All You Need Revisited: Efficient Transformer Architectures for Long-Document Understanding

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## Abstract

Transformers have become the de facto architecture for natural language processing, yet their quadratic self-attention complexity limits application to long scientific documents. This paper revisits efficient attention mechanisms — Longformer, BigBird, and FlashAttention-2 — for the task of long-document summarization and question answering over full research papers (8k–32k tokens). Using the arXiv-Long and PubMed-Long benchmarks, we show that sliding-window attention with global tokens retains 97% of full-attention quality while reducing memory by 68%. We also propose Chunked Cross-Attention Fusion (CCAF), a simple method that splits documents into 1024-token segments, encodes them independently, and fuses evidence via cross-attention. CCAF achieves ROUGE-L of 41.2 and QA F1 of 73.6, outperforming Longformer-large by 1.8 points with 40% faster inference. We discuss implications for RAG indexing of PDFs.

**Keywords:** Transformer, Efficient Attention, Long Documents, Summarization, Retrieval-Augmented Generation

## 1. Introduction

The original Transformer [1] relies on self-attention with O(n^2) time and memory, which becomes prohibitive when n exceeds 4096 tokens. Scientific PDFs often contain 10,000 to 20,000 tokens, including sections, tables, and references. Truncation to 512 tokens loses critical evidence located in methods and results sections.

Efficient transformers address this via sparse patterns, kernel approximations, or IO-aware implementations. However, direct comparisons on full-paper understanding remain scarce, especially for non-English or mixed-language corpora. This paper fills the gap with controlled experiments and a new fusion baseline.

## 2. Background and Related Work

**2.1 Sparse Attention. Longformer [2] combines sliding-window local attention with task-specific global tokens. BigBird [3] adds random attention to approximate full attention theoretically. Both reduce complexity to O(n).**

**2.2 IO-Aware Attention. FlashAttention [4] and FlashAttention-2 reorder computation to minimize HBM reads/writes without approximating attention. It enables 4x longer contexts on the same GPU.**

**2.3 Long-Document Benchmarks. Datasets such as SCITLDR, arXiv summarization, and QASPER [5] test models on full papers. Retrieval-augmented pipelines often chunk papers before encoding, which motivates our fusion approach.**

## 3. Proposed Method: Chunked Cross-Attention Fusion

CCAF operates in three steps. First, a document of length L is split into m chunks of 1024 tokens with 128-token overlap to preserve sentence boundaries. Second, each chunk is encoded independently with a frozen E5 or BERT encoder to produce token representations. Third, a lightweight 2-layer cross-attention decoder attends over concatenated chunk representations conditioned on the query or summarization prompt.

| Architecture | Memory (GB) | ROUGE-L | QA F1 | Speed (doc/s) |
|---|---|---|---|---|
| Full Attention (512 trunc) | 11.2 | 35.4 | 62.1 | 2.4 |
| Longformer-large (16k) | 18.6 | 39.4 | 71.8 | 1.1 |
| BigBird (8k) | 16.3 | 38.7 | 70.5 | 1.3 |
| CCAF (ours, 32k) | 9.8 | 41.2 | 73.6 | 1.9 |

This design decouples encoding from fusion, allowing pre-computed chunk embeddings to be reused in RAG vector stores such as FAISS or Chroma. It is therefore directly compatible with BasicRAG ingestion workflows.

## 4. Experiments

**4.1 Setup. We evaluate on arXiv-Long (10,000 papers, avg. 14k tokens) and PubMed-Long (8,000 papers, avg. 11k tokens). Metrics are ROUGE-L for summarization and token F1 for QA. All models are trained on 4xA100 GPUs with mixed precision.**

**4.2 Results. As shown in Table 1, truncation baselines perform worst, confirming information loss. Longformer and BigBird improve substantially but require high memory. CCAF achieves the best quality while using only 9.8 GB, because chunk encodings are computed in parallel and fusion attends over compressed CLS vectors.**

**4.3 Analysis. Attention visualization reveals that global tokens in Longformer focus on abstract and conclusion sentences, while CCAF cross-attention distributes weight to methods and tables — crucial for scientific QA. Ablation shows overlap of 128 tokens adds +0.7 ROUGE-L over non-overlapping splits. For RAG practitioners, we recommend 512–1024 token chunks for indexing full PDFs.**

## 5. Conclusion

Efficient attention makes full-paper understanding feasible on a single GPU. Our proposed CCAF method offers a simple, fast, and RAG-compatible alternative to sparse transformers. In future work we will test CCAF on Vietnamese scientific papers and scanned PDFs requiring OCR. We release code and checkpoints to support further research on long-document RAG.

## References


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